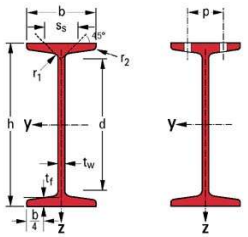
# Avrupa kesit standart profiller

# IPN

flange slope: 14%




| | KESİT | ÖLÇÜLER | | | | | | KESİT ALANI | TASARIM ÖLÇÜLERİ | | | | YÜZEY ALANI | | KESİT | STATİK DEĞERLER | | | | | | | | | | | | SINIFLAMA ENV 1993 1-1 | | | | yarım profillerde | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | KUVVETLİ EKSEN y-y | | | | | ZAYIF EKSEN z-z | | | | | | | | | | | | | |
| | | G | h | b | $t_w$ | $t_f$ | $r_1$ | $r_2$ | A | d | Ø | Pmin | Pmax | $A_L$ | $A_G$ | | G | $I_y$ | $W_{el.y}$ | $W_{pl.y}$ | $i_y$ | $A_{vz}$ | $I_z$ | $W_{el.z}$ | $W_{pl.z}$ | $i_z$ | $s_s$ | $I_t$ | $I_w x10^{-3}$ | EĞİLME | | BASINÇ | | yg | J | Cw |
| | | kg/m | mm | mm | mm | mm | mm | mm | $cm^2$ | mm | | mm | mm | $m^2/m$ | $m^2/t$ | | kg/m | $cm^4$ | $cm^3$ | $cm^3$ | cm | $cm^2$ | $cm^4$ | $cm^3$ | $cm^3$ | cm | mm | $cm^4$ | $cm^6$ | 235 | 355 | 235 | 355 | mm | $cm^4$ | $cm^6$ |
| 1 | IPN 80 | 5.94 | 80 | 42 | 3.9 | 5.9 | 3.9 | 2.3 | 7.57 | 59 | | | | | | | 5.94 | 77.8 | 19.5 | 22.8 | 3.21 | | 6.29 | 3 | 5 | 0.912 | 21.6 | 0.87 | 0.09 | | | | | 9.9 | 0.4 | 0.1 |
| 2 | IPN 100 | 8,3 | 100 | 50 | 4.5 | 6.8 | 4.5 | 2.7 | 10.6 | 75.7 | - | - | - | 0.370 | 44.47 | | 8.3 | 171 | 34.2 | 39.8 | 4.01 | 4.85 | 12.2 | 4.9 | 8.1 | 1.07 | 25.0 | 1.6 | 0.27 | | | | | 12.5 | 0.7 | 0.3 |
| 3 | IPN 120 | 11.1 | 120 | 58 | 5.1 | 7.7 | 5.1 | 3.1 | 14.2 | 92.4 | - | - | - | 0.439 | 39.38 | IPN 120 | 11.1 | 328 | 54.7 | 63.6 | 4.81 | 6.63 | 21.5 | 7.41 | 12.4 | 1.23 | 28.4 | 2.71 | 0.69 | 1 | 1 | 1 | 1 | 15.2 | 1.1 | 0.6 |
| 4 | IPN 140 | 14.3 | 140 | 66 | 5.7 | 8.6 | 5.7 | 3.4 | 18.3 | 109.1 | - | - | - | 0.502 | 34.94 | IPN 140 | 14.3 | 573 | 81.9 | 95.4 | 5.61 | 8.65 | 35.2 | 10.7 | 17.9 | 1.40 | 31.8 | 4.32 | 1.54 | 1 | 1 | 1 | 1 | 17.9 | 1.8 | 1.3 |
| 5 | IPN 160 | 17.9 | 160 | 74 | 6.3 | 9.5 | 6.3 | 3.8 | 22.8 | 125.8 | - | - | - | 0.575 | 32.13 | IPN 160 | 17.9 | 935 | 117 | 136 | 6.4 | 10.83 | 54.7 | 14.8 | 24.9 | 1.55 | 35.2 | 6.57 | 3.14 | 1 | 1 | 1 | 1 | 20.4 | 2.7 | 2.4 |
| 6 | IPN 180 | 21.9 | 180 | 82 | 6.9 | 10.4 | 6.9 | 4.1 | 27.9 | 142.4 | - | - | - | 0.640 | 29.22 | IPN 180 | 21.9 | 1450 | 161 | 187 | 7.2 | 13.35 | 81.3 | 19.8 | 33.2 | 1.71 | 38.6 | 9.58 | 5.92 | 1 | 1 | 1 | 1 | 23.0 | 4.0 | 4.3 |
| 8 | IPN 200 | 26.2 | 200 | 90 | 7.5 | 11.3 | 7.5 | 4.5 | 33.4 | 159.1 | - | - | - | 0.709 | 27.04 | IPN 200 | 26.2 | 2140 | 214 | 250 | 8.0 | 16.03 | 117 | 26.0 | 43.5 | 1.87 | 42.0 | 13.5 | 10.5 | 1 | 1 | 1 | 1 | 25.1 | 5.7 | 7.3 |
| 9 | IPN 220 | 31.1 | 220 | 98 | 8.1 | 12.2 | 8.1 | 4.9 | 39.5 | 175.8 | M10 | 50 | 56 | 0.775 | 24.99 | IPN 220 | 31.1 | 3060 | 278 | 324 | 8.8 | 19.06 | 162 | 33.1 | 55.7 | 2.02 | 45.4 | 18.6 | 17.8 | 1 | 1 | 1 | 1 | 28.0 | 7.8 | 11.9 |
| 10 | IPN 240 | 36.2 | 240 | 106 | 8.7 | 13.1 | 8.7 | 5.2 | 46.1 | 192.5 | M10 | 54 | 60 | 0.844 | 23.32 | IPN 240 | 36.2 | 4250 | 354 | 412 | 9.59 | 22.33 | 221 | 41.7 | 70.0 | 2.20 | 48.9 | 25.0 | 28.7 | 1 | 1 | 1 | 1 | 30.6 | 10.4 | 18.6 |
| 11 | IPN 260 | 41.9 | 260 | 113 | 9.4 | 14.1 | 9.4 | 5.6 | 53.3 | 208.9 | M12 | 62 | 62 | 0.906 | 21.65 | IPN 260 | 41.9 | 5740 | 442 | 514 | 10.4 | 26.08 | 288 | 51.0 | 85.9 | 2.32 | 52.6 | 33.5 | 44.1 | 1 | 1 | 1 | 1 | 33.6 | 14.0 | 28.1 |
| 12 | IPN 280 | 47.9 | 280 | 119 | 10.1 | 15.2 | 10.1 | 6.1 | 61.0 | 225.1 | M12 | 68 | 68 | 0.966 | 20.17 | IPN 280 | 47.9 | 7590 | 542 | 632 | 11.1 | 30.18 | 364 | 61.2 | 103 | 2.45 | 56.4 | 44.2 | 64.6 | 1 | 1 | 1 | 1 | 36.4 | 18.5 | 41.1 |
| 14 | IPN 300 | 54.2 | 300 | 125 | 10.8 | 16.2 | 10.8 | 6.5 | 69.0 | 241.6 | M12 | 70 | 74 | 1.03 | 19.02 | IPN 300 | 54.2 | 9800 | 653 | 762 | 11.9 | 34.58 | 451 | 72.2 | 121 | 2.56 | 60.1 | 56.8 | 91.8 | 1 | 1 | 1 | 1 | 39.6 | 23.7 | 57.7 |
| 15 | IPN 320 | 61.0 | 320 | 131 | 11.5 | 17.3 | 11.5 | 6.9 | 77.7 | 257.9 | M12 | 70 | 80 | 1.09 | 17.87 | IPN 320 | 61.0 | 12510 | 782 | 914 | 12.7 | 39.26 | 555 | 84.7 | 143 | 2.67 | 63.9 | 72.5 | 129 | 1 | 1 | 1 | 1 | 42.4 | 30.3 | 80.8 |
| 16 | IPN 340 | 68.0 | 340 | 137 | 12.2 | 18.3 | 12.2 | 7.3 | 86.7 | 274.3 | M12 | 78 | 86 | 1.15 | 16.90 | IPN 340 | 68.0 | 15700 | 923 | 1080 | 13.5 | 44.27 | 674 | 98.4 | 166 | 2.80 | 67.6 | 90.4 | 176 | 1 | 1 | 1 | 1 | 45.4 | 37.7 | 109.4 |
| 17 | IPN 360 | 76.1 | 360 | 143 | 13 | 19.5 | 13 | 7.8 | 97.0 | 290.2 | M12 | 78 | 92 | 1.21 | 15.89 | IPN 360 | 76.1 | 19610 | 1090 | 1276 | 14.2 | 49.95 | 818 | 114.0 | 194 | 2.90 | 71.8 | 115 | 240 | 1 | 1 | 1 | 1 | 48.5 | 47.8 | 150.6 |
| 18 | IPN 380 | 84.0 | 380 | 149 | 13.7 | 20.5 | 13.7 | 8.2 | 107 | 306.7 | M16 | 84 | 86 | 1.27 | 15.12 | IPN 380 | 84.0 | 24010 | 1260 | 1482 | 15.0 | 55.55 | 975 | 131.0 | 221 | 3.02 | 75.4 | 141 | 319 | 1 | 1 | 1 | 1 | 51.5 | 58.2 | 197.9 |
| 20 | IPN 400 | 92.4 | 400 | 155 | 14.4 | 21.6 | 14.4 | 8.6 | 118 | 322.9 | M16 | 86 | 92 | 1.33 | 14.36 | IPN 400 | 92.4 | 29210 | 1460 | 1714 | 15.7 | 61.69 | 1160 | 149 | 253 | 3.13 | 79.3 | 170 | 420 | 1 | 1 | 1 | 1 | 54.7 | 70.9 | 260.6 |
| 21 | IPN 450 | 115 | 450 | 170 | 16.2 | 24.3 | 16.2 | 9.7 | 147 | 363.6 | M16 | 92 | 106 | 1.48 | 12.83 | IPN 450 | 115 | 45850 | 2040 | 2400 | 17.7 | 77.79 | 1730 | 203 | 345 | 3.43 | 88.9 | 267 | 791 | 1 | 1 | 1 | 1 | 61.7 | 111.5 | 489.6 |
| 23 | IPN 500 | 141 | 500 | 185 | 18 | 27 | 18 | 10.8 | 179 | 404.3 | M20 | 102 | 110 | 1.63 | 11.60 | IPN 500 | 141 | 68740 | 2750 | 3240 | 19.6 | 95.6 | 2480 | 268 | 456 | 3.72 | 98.5 | 402 | 1400 | 1 | 1 | 1 | 1 | 69.0 | 167.4 | 865.5 |
| 24 | IPN 550 | 166 | 550 | 200 | 19 | 30 | 19 | 11.9 | 212 | 445.6 | M22 | 112 | 118 | 1.80 | 10.80 | IPN 550 | 166 | 99180 | 3610 | 4240 | 21.6 | 111.30 | 3490 | 349 | 592 | 4.02 | 107.3 | 544 | 2390 | 1 | 1 | 1 | 1 | 75.0 | 239.4 | 1500.0 |